Fraud Detection

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Table of content

Introduction & Problem

01

Introduction

- Online financial transactions are increasing rapidly worldwide.
- Financial fraud is becoming a major cybersecurity challenge.
- Traditional fraud detection systems are often inefficient.
- AI and machine learning provide smarter fraud detection solutions.

VISION

To create a smart and reliable fraud detection system that enhances financial security using artificial intelligence technologies.

MISION

- Detect fraudulent transactions accurately.
- Reduce financial risks and losses.
- Provide real-time fraud monitoring.

Problem Statement

- Manual monitoring is difficult due to huge transaction volumes.
- False predictions can affect customers and financial institutions.
- Intelligent systems are needed for accurate fraud detection.

Objectives

01

Develop an intelligent fraud detection system using Machine Learning

02

Detect fraudulent credit card transactions accurately in real time.

03

Reduce false positives and improve fraud prediction reliability.

04

Build an interactive Streamlit for transaction monitoring and risk visualization.

Data Analysis & Preparation

02

Dataset

The dataset simulates real-world financial transactions and provides realistic fraud behavior patterns, making it highly suitable for machine learning fraud detection tasks.

Data overview

- Kaggle Fraud Detection Dataset.
- Around 1.8 million transactions
- Contains fraudulent and legitimate transactions

```
Train shape: (1296675, 23)
Test shape:  (555719, 23)
```

trans_date_trans_time	cc_num	merchant	category	amt	first	last	gender	street	...	lat	long	city_pop	
2019-01-01 00:00:18	2703186189652095	fraud_Rippin, Kub and Mann	misc_net	4.97	Jennifer	Banks	F	561 Perry Cove	...	36.0788	-81.1781	3495	Psych cour
2019-01-01 00:00:44	630423337322	fraud_Heller, Gutmann and Zieme	grocery_pos	107.23	Stephanie	Gill	F	43039 Riley Greens Suite 393	...	48.8878	-118.2105	149	educ needs t
2019-01-01 00:00:51	38859492057661	fraud_Lind-Buckridge	entertainment	220.11	Edward	Sanchez	M	594 White Dale Suite 530	...	42.1808	-112.2620	4154	conse
2019-01-01 00:01:16	3534093764340240	fraud_Kutch, Hermiston and Farrell	gas_transport	45.00	Jeremy	White	M	9443 Cynthia Court Apt. 038	...	46.2306	-112.1138	1939	Patent at
2019-01-01 00:03:06	375534208663984	fraud_Keeling-Crist	misc_pos	41.96	Tyler	Garcia	M	408 Bradley Rest	...	38.4207	-79.4629	99	mov psychoth

Data Cleaning & Preprocessing

- Checked for missing values
- Removed irrelevant columns
- Removed duplicate records
- Converted date-time columns
- Encoded categorical variables

Imbalanced Data

	count	percentage
Not Fraud	1289169	99.421135
Fraud	7506	0.578865

- Machine learning models may become biased toward legitimate transactions.
- Detecting fraud cases is more important than maximizing overall accuracy.

Techniques Applied

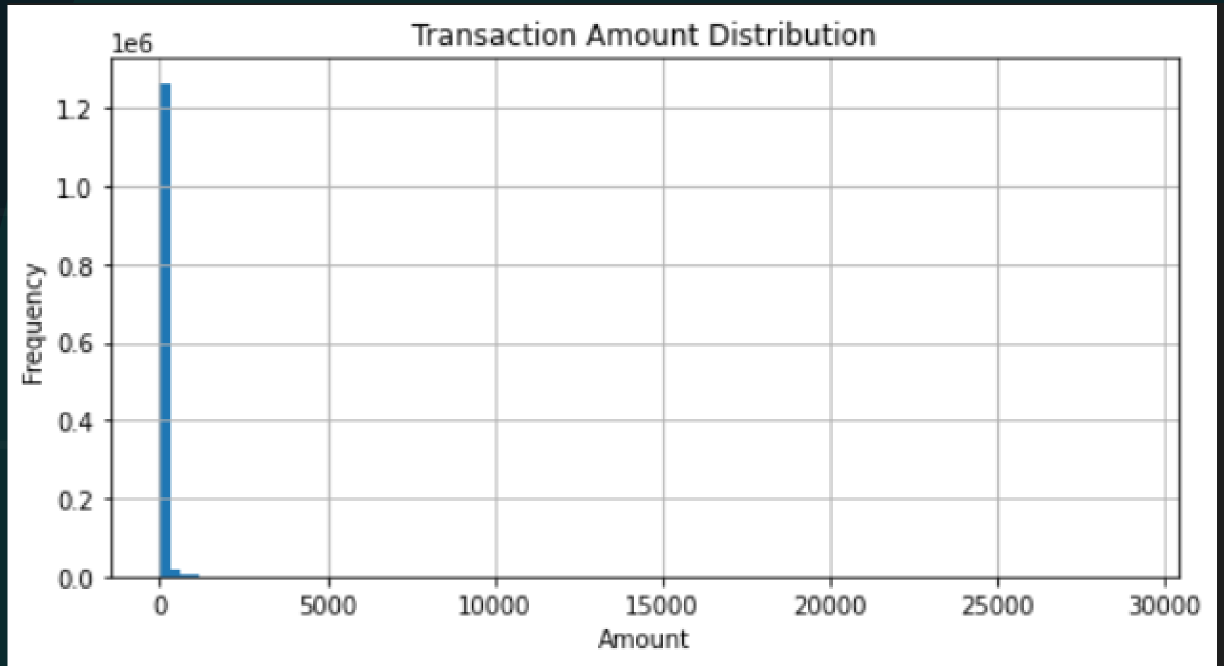
- SMOTE with Logistic Regression
- Threshold optimization

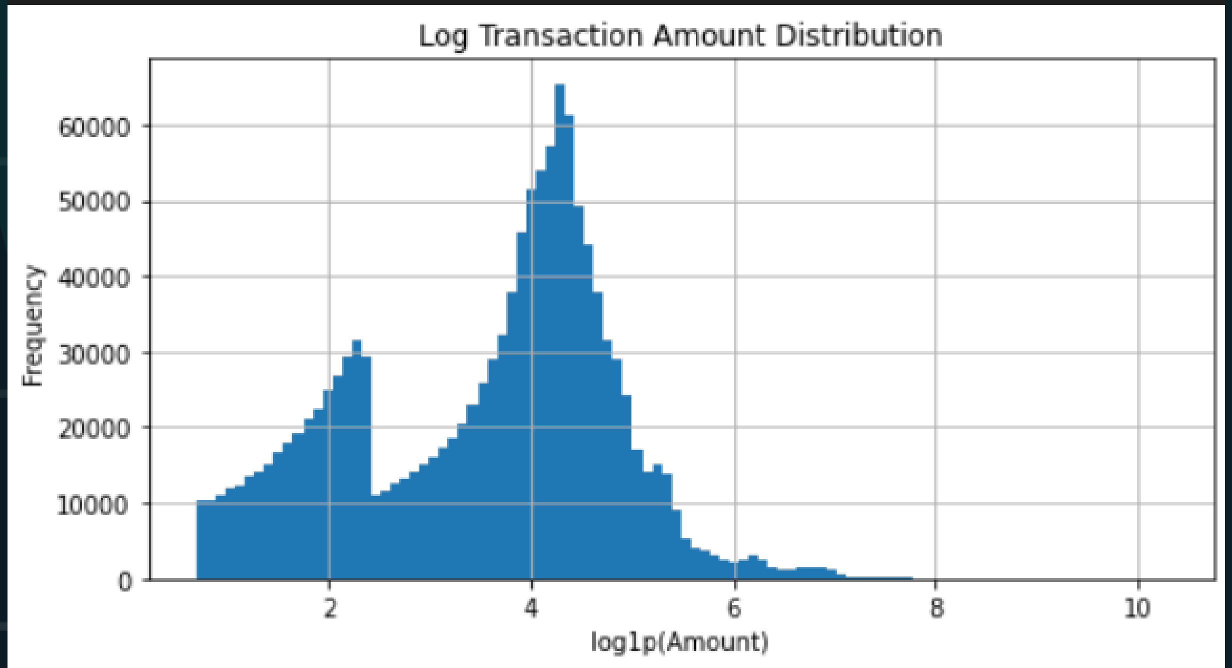
Feature Engineering

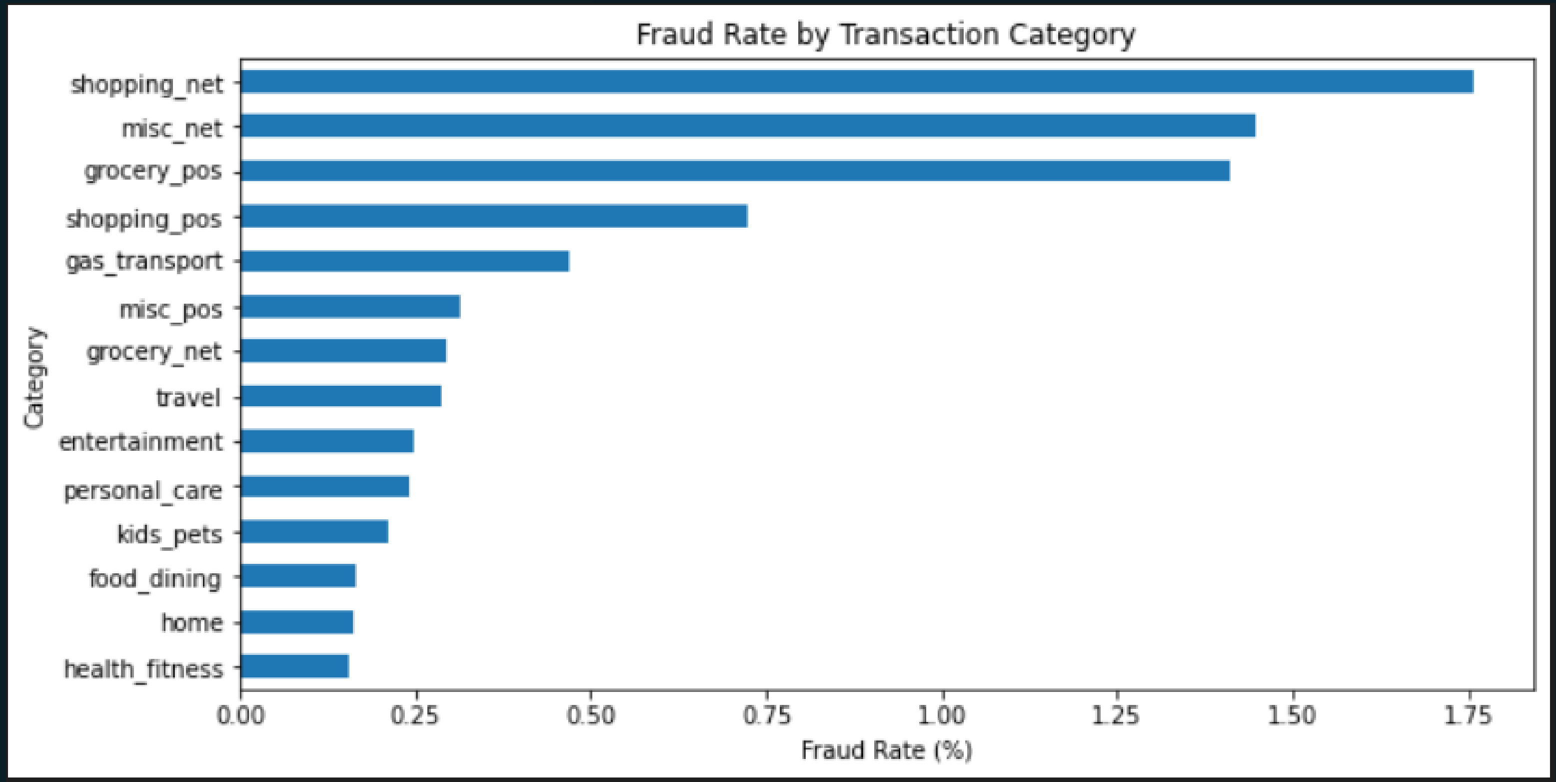
trans_hour	trans_dayofweek	trans_month	trans_day	age	amt_log	distance_km
0	1	1	1	31	1.786747	78.597568
0	1	1	1	41	4.684259	30.212176
0	1	1	1	57	5.398660	108.206083
0	1	1	1	52	3.828641	95.673231
0	1	1	1	33	3.760269	77.556744

- Transaction hour and day
- Customer-merchant distance
- Behavioral transaction indicators
- Risk-related patterns

EDA


Transaction Amount Distribution
1e6
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Frequency
0
5000
10000
15000
20000
25000
30000
Amount


Log Transaction Amount Distribution
60000
50000
40000
30000
20000
10000
0
Frequency
2
4
6
8
10
log1p(Amount)


Fraud Rate by Transaction Category
shopping_net
misc_net
grocery_pos
shopping_pos
gas_transport
misc_pos
grocery_net
travel
entertainment
personal_care
kids_pets
food_dining
home
health_fitness
Category
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
Fraud Rate (%)


Fraud Rate by Gender
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Fraud Rate (%)
F
M
Gender

Project Development

03

Machine Learning Models

```
============================================================
Logistic Regression - Balanced
============================================================
Threshold used: 0.5

Prediction counts:
Not Fraud    518560
Fraud         37159
Name: count, dtype: int64

Classification Report:
              precision    recall  f1-score   support

   Not Fraud       1.00      0.93      0.96    553574
       Fraud       0.02      0.32      0.04      2145

    accuracy                           0.93    555719
   macro avg       0.51      0.63      0.50    555719
weighted avg       0.99      0.93      0.96    555719
```

```
============================================================
Random Forest - Balanced
============================================================
Threshold used: 0.5

Prediction counts:
Not Fraud    552640
Fraud          3079
Name: count, dtype: int64

Classification Report:
              precision    recall  f1-score   support

   Not Fraud       1.00      1.00      1.00    553574
       Fraud       0.57      0.81      0.67      2145

    accuracy                           1.00    555719
   macro avg       0.78      0.91      0.83    555719
weighted avg       1.00      1.00      1.00    555719
```

```
============================================================
XGBoost - Weighted
============================================================
Threshold used: 0.5

Prediction counts:
Not Fraud    547491
Fraud          8228
Name: count, dtype: int64

Classification Report:
              precision    recall  f1-score   support

   Not Fraud       1.00      0.99      0.99    553574
       Fraud       0.25      0.95      0.39      2145

    accuracy                           0.99    555719
   macro avg       0.62      0.97      0.69    555719
weighted avg       1.00      0.99      0.99    555719
```

Handling Imbalance Data

```
=======================================================
Logistic Regression + SMOTE
=======================================================
Threshold used: 0.5

Prediction counts:
Not Fraud    519615
Fraud         36104
Name: count, dtype: int64

Classification Report:
              precision    recall  f1-score   support

   Not Fraud       1.00      0.94      0.97    553574
       Fraud       0.01      0.25      0.03      2145

    accuracy                           0.93    555719
   macro avg       0.51      0.59      0.50    555719
weighted avg       0.99      0.93      0.96    555719
```

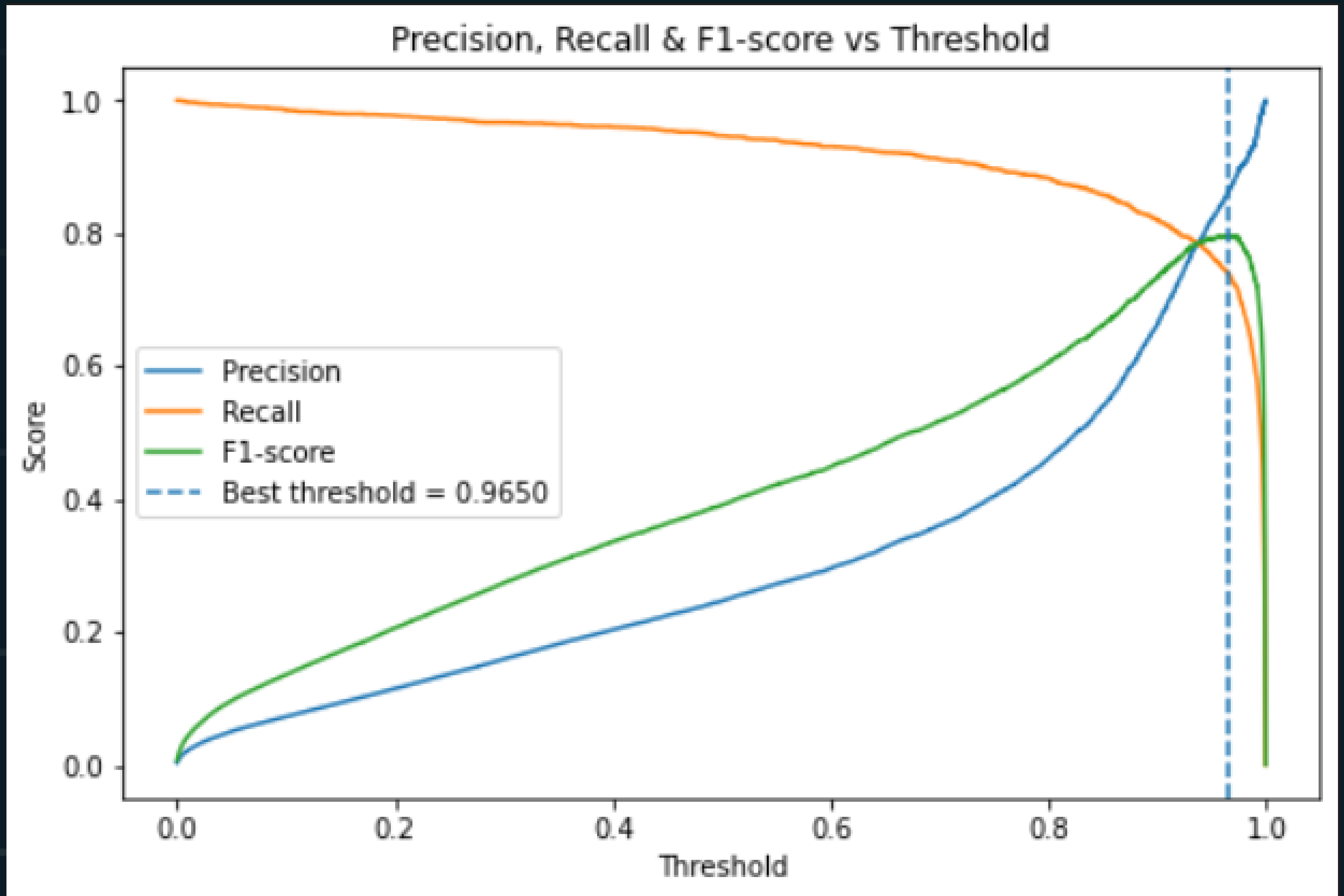
SMOTE was applied with Logistic Regression to handle the imbalanced dataset. Although the model achieved high accuracy, accuracy alone was misleading for fraud detection. Therefore, the evaluation focused more on Precision, Recall, and F1-Score, especially Recall, since detecting actual fraudulent transactions was the main project objective.

Machine Learning Models

```
==========================================================
XGBoost - Weighted
==========================================================
Threshold used: 0.5

Prediction counts:
Not Fraud    547491
Fraud          8228
Name: count, dtype: int64

Classification Report:
              precision    recall  f1-score   support

   Not Fraud       1.00      0.99      0.99    553574
       Fraud       0.25      0.95      0.39      2145

    accuracy                           0.99    555719
   macro avg       0.62      0.97      0.69    555719
weighted avg       1.00      0.99      0.99    555719
```

- After comparing multiple models, XGBoost achieved the best fraud detection performance.
- Different thresholds were tested, and 0.9 provided the best balance between detecting fraud and reducing false positives.

Machine Learning Models




```
============================================================
XGBoost - Weighted - Tuned Threshold
============================================================
Threshold used: 0.9649665951728821

Prediction counts:
Not Fraud    553882
Fraud          1837
Name: count, dtype: int64

Classification Report:
              precision    recall  f1-score   support

   Not Fraud       1.00      1.00      1.00    553574
       Fraud       0.86      0.74      0.80      2145

    accuracy                           1.00    555719
   macro avg       0.93      0.87      0.90    555719
weighted avg       1.00      1.00      1.00    555719
```

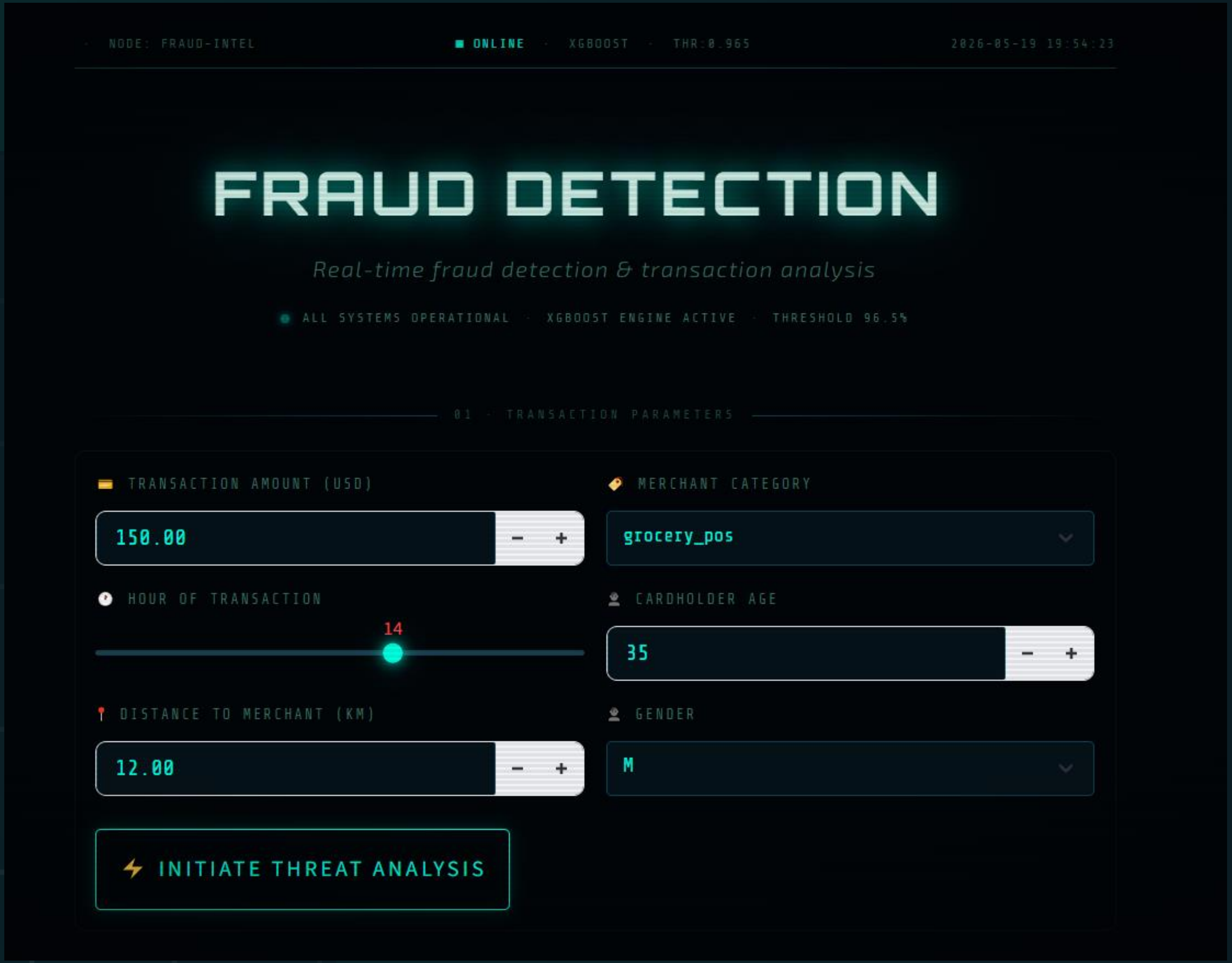
System Design & Implementation

04

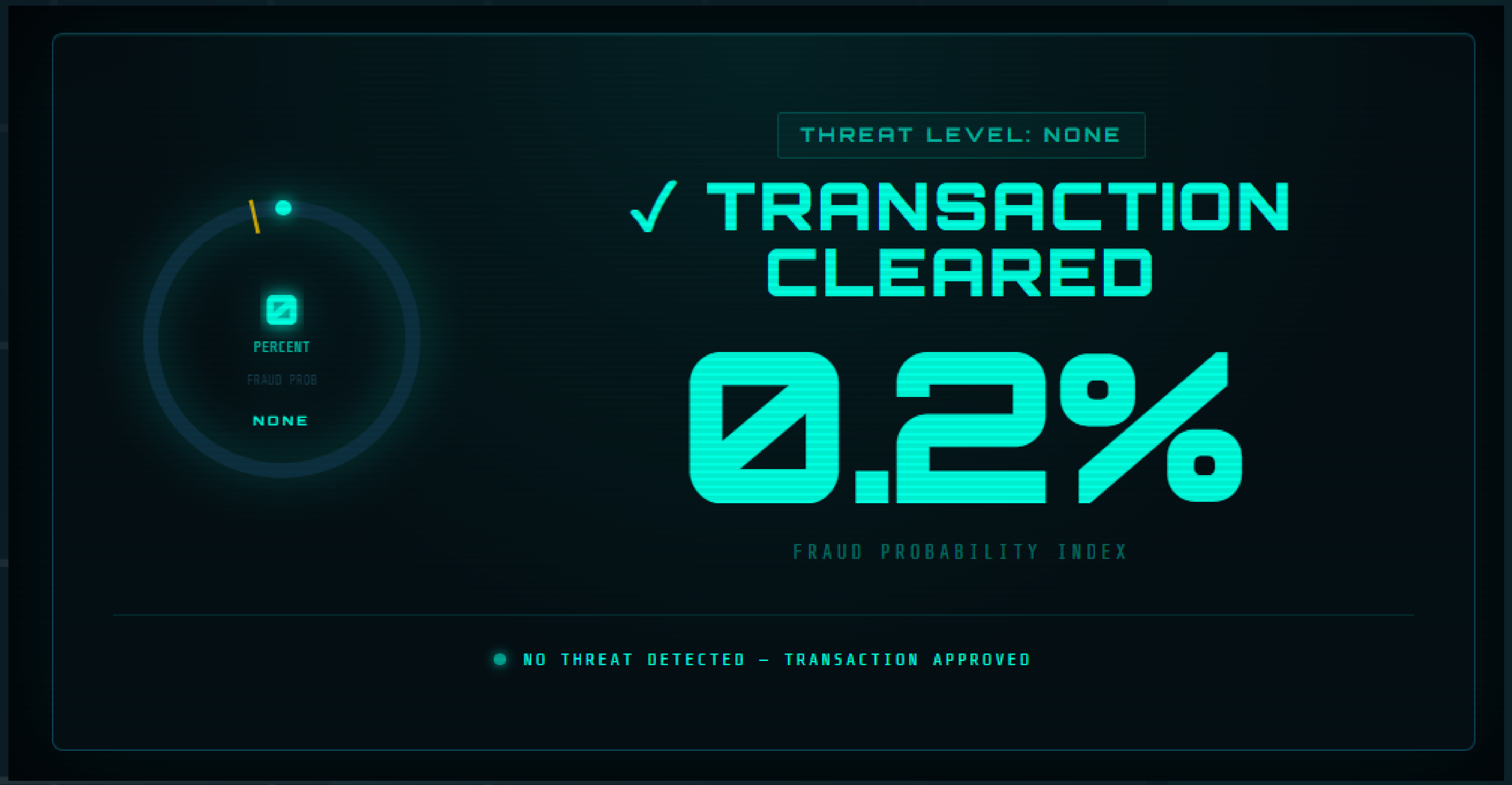
System Architecture

- Transaction Input 01
- Data Processing 02
- XGBoost XGBoost Prediction 03
- Fraud Probability 04
- Dashboard Visualization 05

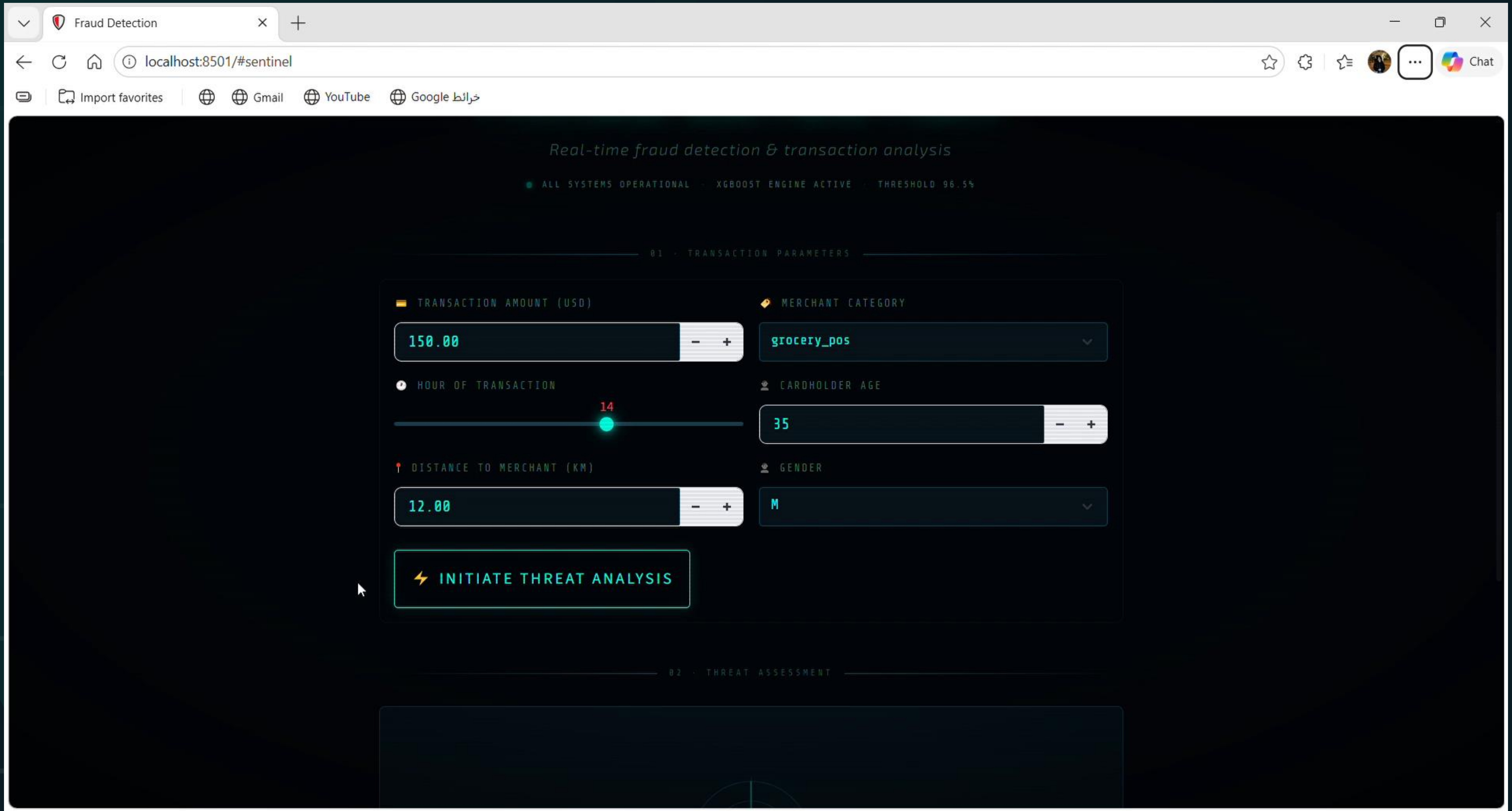
Streamlit Dashboard



Prediction & Risk Analysis



APP DEMO


Fraud Detection
localhost:8501/#sentinel
Import favorites
Gmail
YouTube
Chat
Real-time fraud detection & transaction analysis
ALL SYSTEMS OPERATIONAL · XGBOOST ENGINE ACTIVE · THRESHOLD 96.5%
TRANSACTION AMOUNT (USD)
150.00
MERCHANT CATEGORY
grocery_pos
HOUR OF TRANSACTION
14
CARDHOLDER AGE
35
DISTANCE TO MERCHANT (KM)
12.00
GENDER
M
INITIATE THREAT ANALYSIS

PROJECT EVALUATION&FUTEURE VISION

05

Challenges faced

01

Large Dataset Processing

02

Highly Imbalanced Dataset

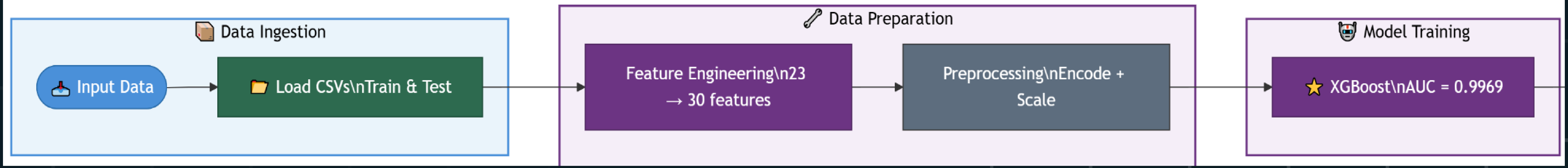
03

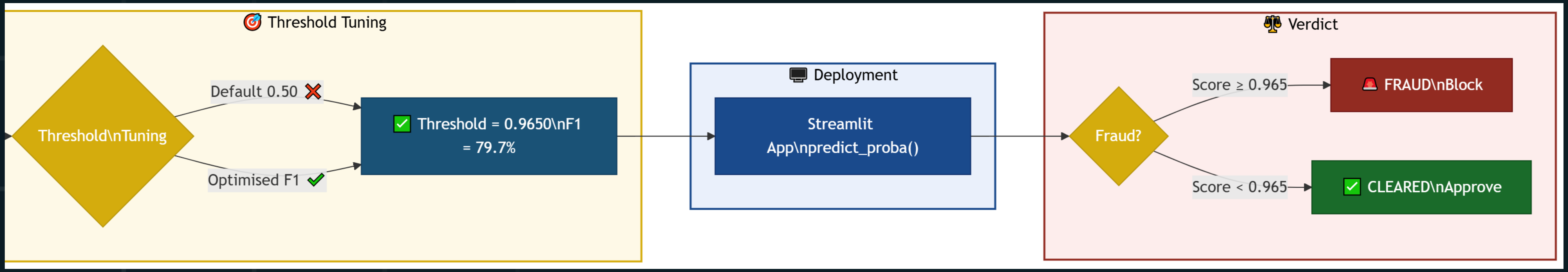
Model Selection & Optimization

Future Improvements

- Cloud deployment
- Banking API Integration
- Deep Learning Enhancement
- Advanced Dashboard Analytics
- Authentication system

CONCLUSION


Data Ingestion
Input Data
Load CSVs\nTrain & Test
Data Preparation
Feature Engineering\n23 → 30 features
Preprocessing\nEncode + Scale
Model Training
XGBoost\nAUC = 0.9969


Threshold Tuning
Threshold\nTuning
Default 0.50
Optimised F1
Threshold = 0.9650\nF1 = 79.7%
Deployment
Streamlit App\npredict_proba()
Verdict
Fraud?
Score ≥ 0.965
Score < 0.965
FRAUD\nBlock
CLEARED\nApprove

QUESTIONS?

GITHUB LINK

STREAMLIT APP

DATASET

Instructor : Sara Hesham